

No. Python Interview Questions

What is Python?

Features of Python?

Python Frameworks?

What is PEP 8?

How to get Id of any object?

a=1, b=1 does both have same Id or not?

Is indentation required in python?

File Extensions in Python?

What is the difference between .py and .pyc files?

What is an Interpreted language?

What is a dynamically typed language?

Python Comments?

What is Docstrings?

Difference between Python comments and docstrings?

What is python Variables?

Ques. Global Variables?

What is Scope in Python?

global Keyword?

What are the common built-in data types in Python?

How We can get the data type of any object?

Primitive/Non-Primitive Data Structures?

What is If Else?

Python While Loops

Switch Statements?

Switch Statements Using class?

Switch Statements Using function?

Type Casting/Type Conversion?

What is Decorators?

What are pickling and unpickling in Python?

No. Python Interview Questions

What is Python JSON?

What is Monkey Patching?

What is Lambda/Anonymous Function?

What is Magic Method Or Dunder Methods?

floor() and ceil() Functions?

try-except-else-and-finally?

Context Manager?

No. Python Operator Interview Questions

What is the operator?

What is membership operator and identity operators?

What is membership operators?

What is Identity operators?

Difference between '==' and 'is' Operator?

No. 	Python List Interview Questions
	What is List?
	Length of the list?
	Check item Exists or not?
	List Comprehension?
	Access List Items/List Slicing?
	Add Items in the List? -- Append, Insert, Extend
	Change or Update List Items?
	Remove List Items?
	-- Remove List Items:- Pop method without index
	-- Remove List Items:- Pop method with index
	-- Remove List Items:- del with keyword
	-- Remove List Items:- del without keyword
	-- Remove List Items:- clear method
	Copy Lists?
	-- Copy List:- Copy Method
	-- Copy List:- list Method
	Join Lists?
	-- Join Lists:- Using plus(+)
	-- Join Lists:- Using append
	-- Join Lists:- Using extend
	Loop Lists?
	Sort Lists?
	flatten list?
	List Methods?
	-- List Methods:- Append
	-- List Methods:- Insert
	-- List Methods:- Extend
	-- List Methods:- Copy
	-- List Methods:- Count
	-- List Methods:- Index

No. Python List Interview Questions**-- List Methods:-** Pop**-- List Methods:-** remove**-- List Methods:-** clear**-- List Methods:-** Reverse**-- List Methods:-** SortDifference between **List** and **Tuples** in Python?

No. Python List Logical Interview Questions

Find the Length of a List?

Length hint function?

Finding length of list using loop?

Interchange first and last elements in a list?

Swap Two Elements in a List?

Swap elements in String list?

find the single number(unique-element) of the list?

find the even number from the list?

Print duplicate list, Find Even Or Odd Number?

Find the duplicate element from list?

Remove duplicate item from list using List comprehension?

Convert a list into string?

Write a program to print a list in reverse order?

find the max, min number from the list user input?

find the sum of list elements?

Generate a number list between two ranges?

Remove elements in a list after a specific index?

Remove elements in a list before a specific index?

Remove elements in a list between 2 indices?

How would you convert a list to an array?

No. Python Tuple Interview Questions

What is Tuples?

Tuple Length?

Create Tuple With One Item?

Check if Item Exists in tuple

Tuple() Constructor?

packing and unpacking Tuples?

Access Tuple Items?

Add Tuples Items?

-- Add Tuples Items:- Append

-- Add Tuples Items:- Insert

-- Add Tuples Items:- Extend

Change or Update Tuples?

Remove Tuples Items?

-- Remove Tuples Items:- Remove

-- Remove Tuples Items:- Pop

-- Remove Tuples Items:- del

-- Remove Tuples Items:- Clear

Copy tuple?

-- Copy tuple:- tuple method

-- Copy tuple:- copy method

-- Copy tuple:- deep copy method

Join Tuples?

-- Join Tuples:- Plus

-- Join Tuples:- Multiply

Loop Tuples?

Loop sort?

Tuple Methods?

-- Tuple Methods:- count

-- Tuple Methods:- index

Difference between List and Tuples in Python?

No. Python Tuple Program Interview Questions

Convert a list into a tuple?

How to Sort List Of Tuples By The First Element?

How to Sort List Of Tuples By The Second Element?

No. Python Set Questions

What is Set?

Get the Length of a Set?

set() Constructor?

Loop Sets?

Acesss Items of set?

Add Items of set?

-- **Add Items:-** Add

-- **Add Items:-** Update

Remove Item of set?

What is difference between Discard() and Remove()?

Join Two Set?

Set Methods?

No. Python Dictionary Interview Questions

What is Dictionaries?

Dictionary Length?

dict() Constructor?

How to Check if Item Exists?

Access Item of Dictionary?

Add Dictionary Items?

Change Or Update Dictionary Items?

Remove Dictionary Items?

-- **Remove Dictionary:-** pop

-- **Remove Dictionary:-** popitem

-- **Remove Dictionary:-** del

-- **Remove Dictionary:-** del method without key name

-- **Remove Dictionary:-** clear

No. Python Dictionary Interview Questions

Copy Dictionaries?

Copy Dictionaries:- copy

Copy Dictionaries:- dict

Loop Dictionaries?

Dictionary Methods?

Nested Dictionaries?

String Interview Questions

No. Python String Interview Questions

Python Strings

Assign String to a Variable

Multiline Strings

Check String

Slicing Strings

Strings Method OR Change/Modify String

-- Strings Method:- capitalize(), count, Upper, Lower

-- Strings Method:- title, replace, strip, lstrip, rstrip split

-- Strings Method:- join, startswith, endswith

-- Strings Method:- isalpha, isdigit, isalnum

When would you use rfind()

String Concatenation

What is an f-string and how do you use it

Format Strings

Escape Characters

How would you confirm that 2 strings have the same identity

find the length / Count number of characters in a string

No. Logical string Questions

Looping Through a String

Extract numbers from string

Reversed the String

No. Logical string Questions

Reversed word in string?

Remove vowels from a string

Find repeated characters in a string python

program to count the frequency of each character**No. Python OOps Interview Questions**

Object-Oriented Programming (OOPS)?

What is the use of **self** in Python?

What is **init** Method?

What is **break**, **continue** and **pass** in Python?

What is the super() Function?

What is **issubclass()**?

What is **_str_** and **_repr_**?

What is **Abstract** Method?

What is **Concrete** Method?

Difference between **method** and **function**?

Accessing the attributes value different ways?

Change the value of the attribute?

What is **_init_** Method?

Delete the Object?

Python Access Modifiers?

What is Python **Enumeration**?

What is Class?

-- Class:- Get Class Variables/attributes

-- Class:- Set/Change values for class variables

-- Class:- Delete class variables

Built-In Class Functions (getattr, setattr, delattr, hasattr) In Python?

What is class meta?

No. Python Oops Interview Questions

What is Object?

Delete the Object?

Counting the Number of objects of a Class?

What is Inheritance?

Single Inheritance

Multiple Inheritance

Multi-Level Inheritance

Hierarchical Inheritance

Hybrid Inheritance

What is Encapsulation?

What is Polymorphism?

What is a constructor

Types of Constructors (Default, Non-parametrized, Parameterized)

Constructor With Default Values?

Constructor Overloading

What is Destructors?

Python Access Modifiers?

private attributes and method?

What is the super() Function

MRO(Method Resolution Order) / Diamond Problem?

What is Abstract Class?

When use abstratc class?

How to Create an Abstract Method

Built-In Class Attributes?

What is Method Overloading?

No. Python Oops Interview Questions

What is Method Overriding?

What is Method?

What is static method?

Python Programs Questions**No. Python Programs Questions**

1 Hello World

2 swap two variables

3 check if a number is Even or odd

4 Check Prime Number Or Not?

5 Prime Number Print between lower to upper?

To Check if a String is a Palindrome

To Check if a Number is a Palindrome

Find the Factorial of a Number

find Fibonacci series up to n

Reverse string?

Reverse a Number?

check number is an Armstrong number or not

No. Python array Programs Questions

Find the missing number in the array?

No. Math Module

what is Math Module?

math.floor() Method

math.ceil() Method

No. Python Other Questions

What is Generator Functions?

What is *args and **kwargs in Python?

What do *(single asterisk) and **(double asterisk)

Why use *args and **kwargs in Python?

Difference between *args and **kwargs in Python?

No. Python Other Questions

[Positional \(non-keyword\) arguments](#)

[Keyword arguments](#)

[Arbitrary Arguments](#)

[What is an Iterable?](#)

[What is Iterators?](#)

[Difference between Iterators and iterable?](#)

[Difference between generator and iterators in python?](#)**No. Python Other Questions**

[Multithreading](#)

Python Interview Questions

2. What are modules and packages in Python?
3. What are global, protected and private attributes in Python?
4. What is the use of self in Python?
5. What is **init**?
6. What is docstring in Python?
7. What is slicing in Python?
8. Explain how can you make a Python Script executable on Unix?
9. What is the difference between Python Arrays and lists?
10. How is memory managed in Python?
11. What are Python namespaces? Why are they used?
12. What is Scope Resolution in Python?
13. What are Dict and List comprehensions?
14. What is lambda in Python? Why is it used?
15. How do you copy an object in Python?
16. What is the difference between xrange and range in Python?
17. What are generators in Python?
18. What is PYTHONPATH in Python?
19. What is the use of help() and dir() functions?
20. What is the difference between .py and .pyc files?
21. How Python is interpreted?
22. How are arguments passed by value or by reference in python?
23. What are iterators in Python?
24. Explain how to delete a file in Python?
25. Explain split() and join() functions in Python?
26. What are negative indexes and why are they used?

Python OOPS Interview Questions

1. How do you create a class in Python?

2. How does inheritance work in python? Explain it with an example.
3. How do you access parent members in the child class?
4. Are access specifiers used in python?
5. Is it possible to call parent class without its instance creation?
6. How is an empty class created in python?
7. Differentiate between new and override modifiers.
8. Why is finalize used?
9. What is init method in python?
10. How will you check if a class is a child of another class?
- 11.

Python Libraries Interview Questions

1. Differentiate between a package and a module in python.
2. Ques:-What is the Python "with" statement designed for?

Ques. What are global, protected and private attributes in Python?

- **Global variables** are public variables that are defined in the global scope. To use the variable in the global scope inside a function, we use the **global** keyword.
- **Protected attributes** are attributes defined with an **single underscore** prefixed to their identifier eg. `_mohit`. They can still be accessed and modified from outside the class they are defined in but a responsible developer should refrain from doing so.
- **Private attributes** are attributes with **double underscore** prefixed to their identifier eg. `__mohit`. They cannot be accessed or modified from the outside directly and will result in an `AttributeError` if such an attempt is made.

Python Collections (Arrays)?

There are four collection data types in the Python programming language:

- **List** is a collection which is ordered and changeable. Allows duplicate members.
- **Tuple** is a collection which is ordered and unchangeable. Allows duplicate members.
- **Set** is a collection which is unordered and unindexed. No duplicate members.
- **Dictionary** is a collection which is unordered, changeable and indexed. No duplicate members.

Ques. What is built-in module in Python?

<https://docs.python.org/3/py-modindex.html>

Example

```
>>> import html
>>> import html.parser
import mysql.connector
```

ye buil in packege hote hai
agar or karne hai to pip ki help sa karenge

Ques. How is memory managed in Python?

- Memory management in Python is handled by the **Python Memory Manager**.
- Python also has an inbuilt garbage collector, which recycles all the unused memory and so that it can be made available to the heap space.
- Memory management in python is managed by Python private heap space. All Python objects and data structures are located in a private heap. The programmer does not have access to this private heap. The python interpreter takes care of this instead.
- The allocation of heap space for Python objects is done by Python's memory manager. The core API gives access to some tools for the programmer to code.

Ques. What is Map, Filter, Reduce?

sdafasf fasf das f

String to array:-

```
thislist = ["apple", "banana", "cherry"]
str1 = ' '.join(thislist)
counter = dict.fromkeys(str1, 0)
print(counter)
for item in str1:
    counter[item] += 1
print(counter)

import collections
print(collections.Counter("hello"))
```

Use of "get()" function

Ques. What is difference betweenr repr() and str()?

the built-in str() and repr() functions both produce a textual representation of an object.

- **str():**- The str() function returns a user-friendly description of an object.
- **repr():**- The repr() method returns a developer-friendly string representation of an object.

```
import datetime
today = datetime.datetime.now()
print(str(today))
print(repr(today))
```

Output:-

```
2021-08-14 08:18:25.138663
datetime.datetime(2021, 8, 14, 8, 18, 25, 138663)
```

```
class Fruit:
    def __init__(self, name):
        self.name = name

banana = Fruit("Banana")
print(banana)
```

Output:- <__main__.Fruit object at 0x7f97c77704c0>

```
-----
class Fruit:
    def __init__(self, name):
        self.name = name

    def __str__(self):
        return f'I am a {self.name}'

banana = Fruit("Banana")
print(banana)
```

Output:- I am a Banana

Ques. What is Scope Resolution in Python?

- scope resolution in python follows the LEGB rules.
 1. Local(L): Defined inside function/class
 2. Enclosed(E): Defined inside enclosing functions(Nested function concept)
 3. Global(G): Defined at the uppermost level
 4. Built-in(B): Reserved names in Python builtin modules
- Local, Enclosed, and Global Scopes in Python

```
pi = 'global pi variable'

def outer():
    pi = 'outer pi variable'
    def inner():
        # pi = 'inner pi variable'
        nonlocal pi
        print(pi)
    inner()

outer()
print(pi)
```

Output:-
outer pi variable
global pi variable

What are Python namespaces?

A namespace in python refers to the name which is assigned to each object in python. The objects are variables and functions. As each object is created, its name along with space (the address of the outer function in which the object is), gets created. The namespaces are maintained in python like a dictionary where the key is the namespace and value is the address of the object. There are 4 types of namespace in python-

Built-in namespace- These namespaces contain all the built-in objects in python and are available whenever python is running. Global namespace- These are namespaces for all the objects created at the level of the main program. Enclosing namespaces- These namespaces are at the higher level or outer function. Local namespaces- These namespaces are at the local or inner function.

```
a = "mohit kumar"
new_list = [1,2,3,4,3,2,1,5,6,7]
list4 = []
for item in a:
    list4.append(item)
print(list4)
list5 = []
[list5.append(item*5) for item in new_list if item not in list5]
print(list5)
dict1 = {}
# for item in List:
#     dict1[len(item)] = item
# print(dict1)
dict1 = {len(item): item for item in List}
print(dict1)
abc = sorted(dict1)
list6 = [dict1[item] for item in abc]
print(list6)
```

Unique dictionary from list?

```
a = [
    {'id':1,'name':'john', 'age':34},
    {'id':1,'name':'john', 'age':34},
    {'id':2,'name':'hanna', 'age':30},
]

b = {x['id']:x for x in a}.values()

print(b)

Output:-
dict_values([{'id': 1, 'name': 'john', 'age': 34}, {'id': 2, 'name': 'hanna', 'age': 30}])
```

Ques. How do you open and close a file in Python?


```
file = open("example.txt", "r") # Opens the file in read mode
content = file.read() # Read the content of the file
print(content)
file.close() # Close the file when done
```

g. WAP to count from a string? l. Count occurrence of each number in list. s. Write a custom insert() function for list eg. def custom_insert (list, index, item). bb. Can we pass list or tuple as a key in dictionary? ee. What is property decorator? ii. What is lazy evaluation in python? b. Static file contains which type of file normally? c. How can be file read in specific location? d. How do you remove a file from a folder? e. If user upload excel file check file format if it is valid or invalid. g. What is Garbage Collector? g. What is the difference between Static method and Class method? h. Private variable and how we can access that? j. Difference multilevel and multiple inheritance and drawback?

1. Exception Handling a. How can we handle errors in python? b. How many ways exception in python?
2. Multithreading a. What is GIL?
3. Numpy a. Tell me some python libraries .. Numpy b. Convert a list into numpy

<https://pynative.com/python-constructors/>

chatgpt Certainly, here are some Python interview questions across different levels of complexity:

Beginner:

How can you iterate over a dictionary?

Intermediate:

Explain the Global Interpreter Lock (GIL) in Python. How does it affect multi-threading? What are list comprehensions? Provide an example. What is the purpose of the **init** method in a Python class? How does exception handling work in Python? Provide examples of using try, except, finally.

Advanced:

What are generators and iterators in Python? How are they different from each other? Explain the concept of "duck typing" in Python. How can you profile and optimize Python code for performance? Describe the use of context managers using the with statement. Explain the concept of metaclasses in Python. What are async functions in Python? How do they relate to asynchronous programming?

Problem-Solving:

Write a function to check if a given string is a palindrome. Implement a binary search algorithm. Given a list of numbers, write a function to find the two numbers that add up to a specific target sum. Implement a basic LRU (Least Recently Used) cache. Remember, interview questions can vary widely based on the specific role and company. It's important to not only memorize answers but also to understand the underlying concepts. Practice coding these questions and explaining your thought process out loud, as interviews often focus on problem-solving skills and how well you can communicate your solutions. #####

AbstractUser vs AbstractBaseUser The default User model in Django uses a username to uniquely identify a user during authentication. If you'd rather use an email address, you'll need to create a custom User model by

either subclassing `AbstractUser` or `AbstractBaseUser`.

Options:

`AbstractUser`: Use this option if you are happy with the existing fields on the `User` model and just want to remove the `username` field. `AbstractBaseUser`: Use this option if you want to start from scratch by creating your own, completely new `User` model.

https://books.agiliq.com/projects/django-orm-cookbook/en/latest/null_vs_blank.html

Python interview questions

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	How to check What type of datatype?
	Python Strings?
	What is built-in module in Python?
	How is memory managed in Python?
	Global Keyword?
	Python Collections (Arrays)?
	Convert a list into a tuple
	Combine two dictionary adding values for common keys?

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Ques. What are global, protected and private attributes in Python?

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Ques. How would you convert a list to an array?

- Python list is a linear data structure that can hold heterogeneous elements. Python does not have a built-in array data type. If you want to create an array in Python, then use the numpy library.
- To *install numpy* in your system, type the following command.
- `python3 -m pip install numpy`
- **1. Using `numpy.array()`**

```
import numpy as np

elon_list = [11, 21, 19, 18, 29]
elon_array = np.array(elon_list)

print(elon_array)
print(type(elon_array))
```

Output:-

```
[11 21 19 18 29]
<class 'numpy.ndarray'>
```

2. Using `numpy.asarray()`

```
import numpy as np
```

```
elon_list = [11, 21, 19, 18, 29]
elon_array = np.asarray(elon_list)

print(elon_array)
print(type(elon_array))
```

Output:- [11 21 19 18 29]

```
<class 'numpy.ndarray'>
```

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```
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counter = dict.fromkeys(str1, 0)
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for item in str1:
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Use of "get()" function

Ques. Why is NumPy Faster Than Lists?

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print(list5)
dict1 = {}
# for item in List:
#     dict1[len(item)] = item
# print(dict1)
dict1 = {len(item): item for item in List}
print(dict1)
abc = sorted(dict1)
list6 = [dict1[item] for item in abc]
print(list6)
```

f

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